Lecture 8: Flexible demand

Energy scarcity, grid scarcity, and the role of price signals

Janek V. Eskildsen[†]

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[†]Department of Economics, UCPH

Outline of today

1. Electrification and demand flexibility

2. The economics of real-time electricity pricing

   The model

   Optimal energy mix

   Consumption and carbon emissions

   The social benefit of reactive consumers

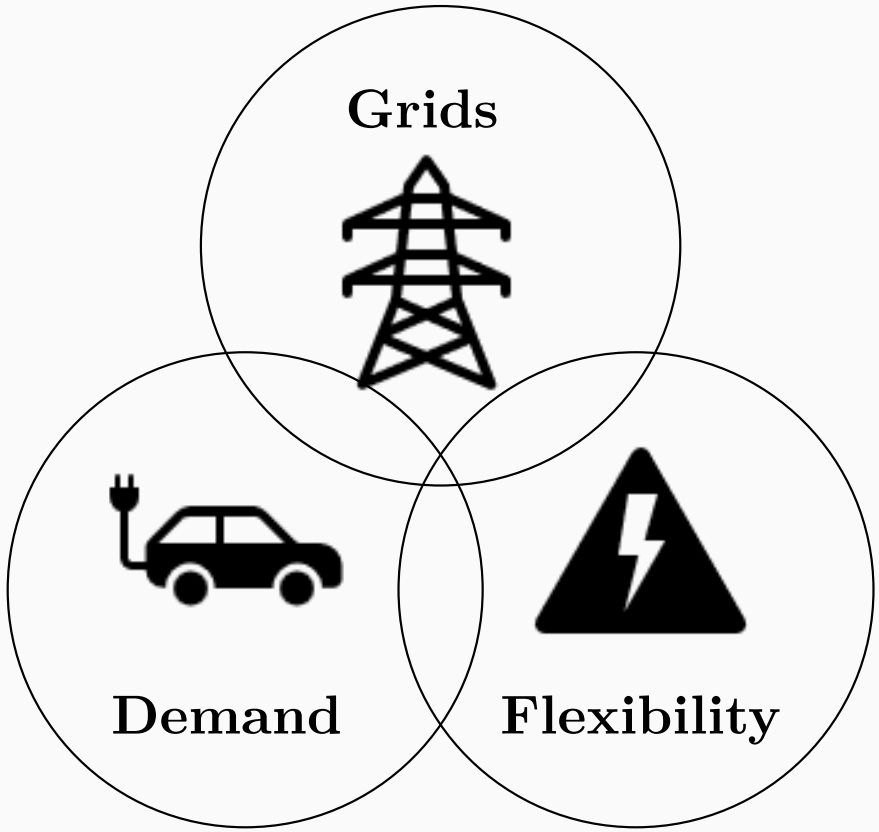
3. Does real-time pricing provide more flexibility?

4. Relieving the grid: flexible demand and network pricing

Electrification and demand flexibility

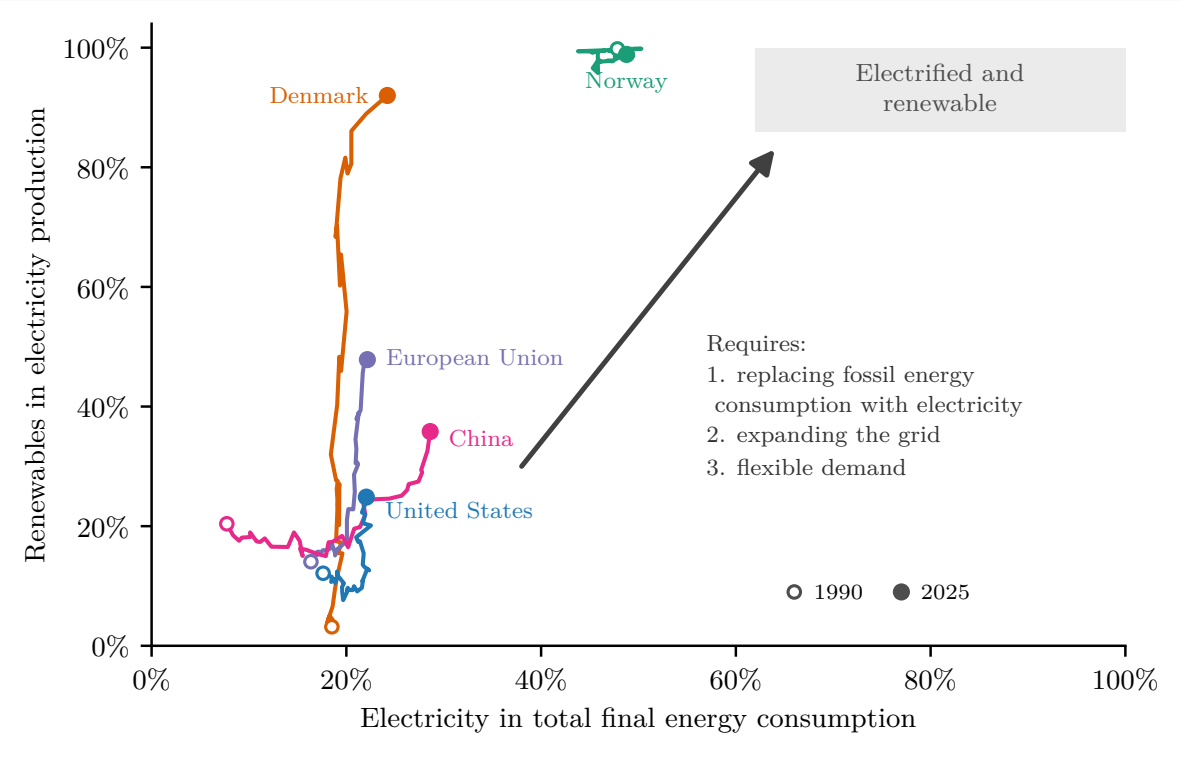
The three elements of electrification

- **Demand:** mainly the *extensive margin*, i.e. adoption of electric technologies (more to come).
- **Grids:** capacity adequacy and congestion management.
- **Flexibility:** trade, storage or price-responsive demand.



Renewable power has grown, electrification has not

So far in the course: the supply side. Now: the demand side.



Source: Enerdata (2026) Energy Statistical Yearbook 2026. Renewables include biomass.

Purpose of today's lecture

The final lecture on flexibility. Today: the demand side.

Flexible demand has two jobs, each with its own price signal:

- **Energy scarcity:** move consumption to hours when energy is abundant. Signal: the spot price.
- **Grid scarcity:** move consumption away from hours when the grid is full. Signal: the network tariff.

We try to answer:

1. How does a larger share of consumers on real-time pricing affect the optimal energy mix and welfare?
2. Does real-time pricing make demand flexible in practice? If not, why not?
3. Do prices and tariffs signal when the grid is full?

The economics of real-time electricity pricing

Correct price signals

- In spot markets, prices signal the scarcity of electricity.
- Time-varying prices should provide the correct incentives to reallocate demand to hours where electricity is abundant.
- However, processing price information and reallocating consumption is not a simple problem for consumers.
- This is particularly the case when consumption decisions have to be taken frequently: Consumers cannot buy electricity in bulk at a low price, store it, and use it when needed.
- For a long time, consumers have been offered, and have accepted, fixed-price contracts: the price is the same regardless of scarcity.

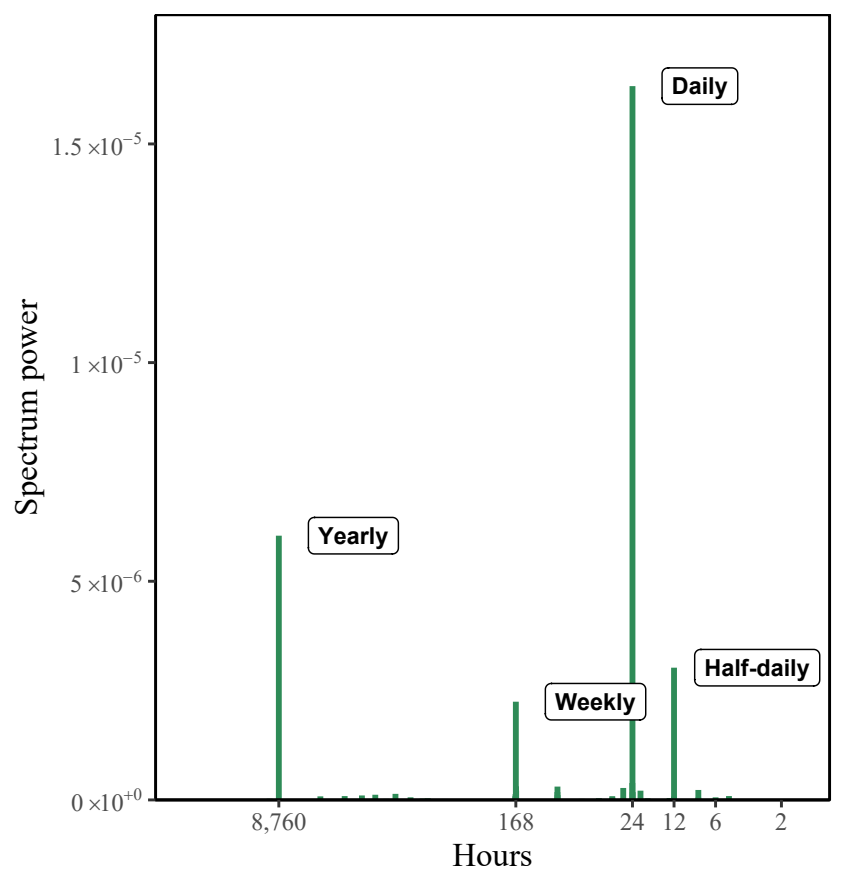
Types of time-varying price schedules

- Real-time pricing (**RTP**): theoretically optimal in neoclassical economics.
- Time-of-Use (**TOU**) pricing: prices follow a fixed daily schedule (e.g. peak vs. off-peak).
- Critical Peak Pricing (**CPP**): a higher price in announced peak hours, with advance notice.
- Voluntary load shedding: payment to large commercial consumers for reducing consumption in peak periods.

Demand historically

Historically: demand has been habit-driven and has not responded to spot prices.

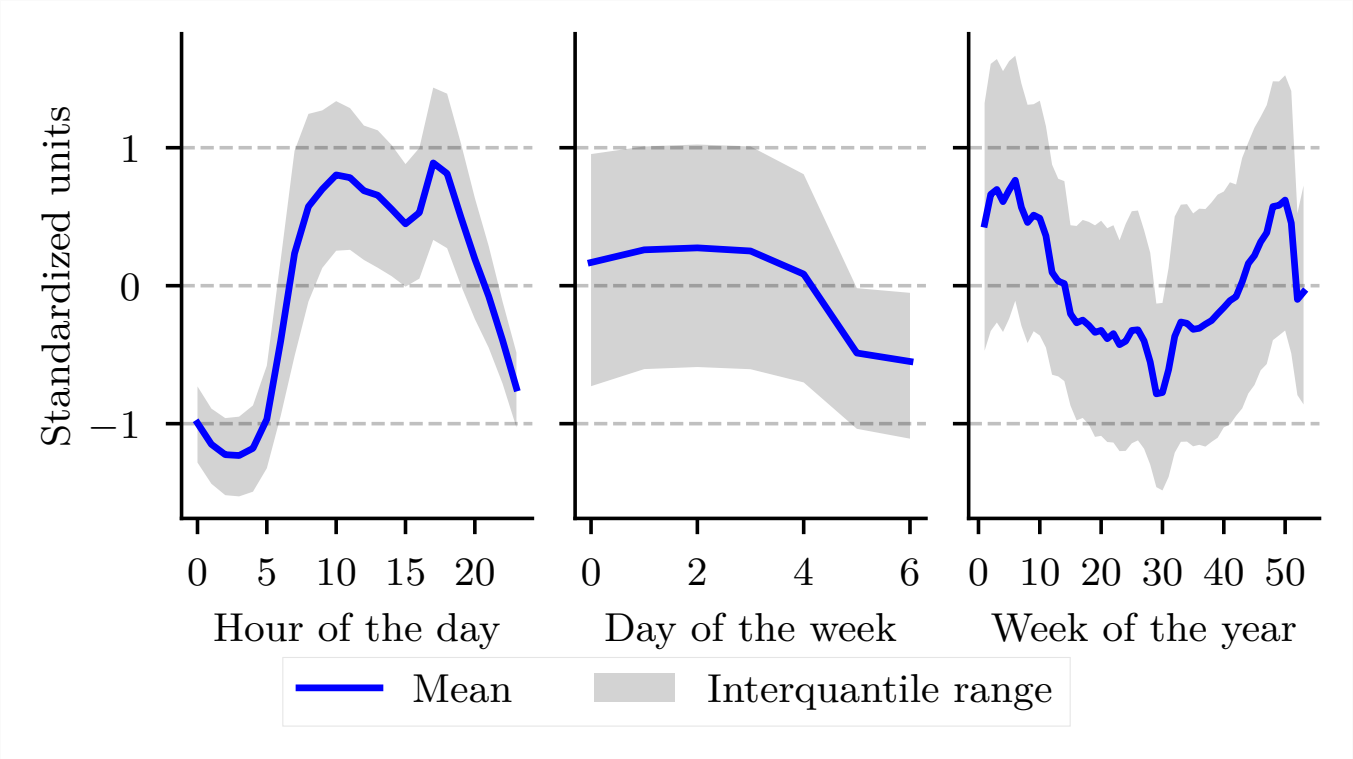
(a) Periodogram, 2011-2017



Demand historically

Historically: demand has been habit-driven and has not responded to spot prices.

Figure 2: Average load in Denmark, 2015-2022



Source: Own calculations based on `www.energidataservice.dk`

New trends provide arguments for RTP

Two new trends challenge the conventional fixed-price contract:

- Digitization: Smart meters make large-scale RTP technologically feasible.
- Intermittency: The case for RTP grows with the share of wind and solar.

RTP is at the heart of energy-market design

RTP is more than a consumer contract:

- On 30 September 2025, the EU day-ahead market moved to 15-minute intervals.



Q: Why is this change important for RTP?

Important assumptions

Ambec and Crampes (2021) study how the introduction of RTP affects the optimal energy mix and derive under what conditions RTP is welfare enhancing.

Their main setup is characterized by:

1. Greenfield investments: long-run equilibrium where current capacities are fully depreciated.
2. Rational consumers.
3. No storage or trade: demand response is the only source of flexibility, which raises its value.

The economics of real-time electricity pricing

The model

The basics

Two markets: A wholesale market and a retail market (both subject to perfect competition).

Two consumer price contracts:

- A share β of consumers pay real-time prices: reactive (r).
- A share $1-\beta$ pay a fixed price: non-reactive ($\overline{r}$).

Two states of nature:

- A windy state $h = w$ occurring with $\Pr(h = w) = \nu$,
- A windless state $h = \overline{w}$ occurring with $\Pr(h = \overline{w}) = (1 - \nu)$.

The wholesale market

On the wholesale market, prices are state dependent and signal the scarcity of electricity:

- p^w is the price in the windy state.
- $p^{\overline{w}}$ is the price in the windless state.

Consumers that face the state-dependent wholesale prices are called reactive and consume q_r^w in state w and $q_r^{\overline{w}}$ in state $\overline{w}$.

The retail market

Retailers purchase power on the wholesale market and provide electricity at a fixed rate p.

Perfect competition on the retail market implies that retailers offer the actuarially fair price:[1]

$$p = \nu p^{w} + (1 - \nu)p^{\overline{w}}.$$

Consumers that face the fixed retail price p are called non-reactive and consume $q_{\overline{r}}$ in both states of nature.

[1] The actuarially fair price is the price at which the retailer makes zero-expected profits.

Production

Electricity is generated by means of two technologies (denoted j):

1. Dispatchable thermal plant ($j = f$):
   - Generation denoted $q_f^h \in [0, K_f]$,
   - Constant unit costs of generation c,
   - Constant unit costs of capacity r_f.
2. Non-dispatchable wind farm reliant on intermittent wind speeds ($j = i$):
   - Generation is $q_i^w \in [0, K_i]$, $q_i^{\overline{w}} = 0$.
   - Unit costs of capacity are $r_i \in [\underline{r}_i, \infty)$, distributed according to pdf $f(r_i)$ and cdf $F(r_i)$.
   - They assume a total potential maximum capacity of $\overline{K}$ for every r_i (i.e. $K_i \leq \overline{K}$)

Optimal investment in intermittent capacity

What does a distribution of wind capacity costs mean?

- Assume there is a continuum of wind turbines each with a fixed capacity of 1 unit (e.g. kW).
- The total mass of wind turbines is $\overline{K}$ (e.g. constrained by locations).
- Each wind turbine has *constant* marginal capacity costs but the capacity costs vary in the range $r_i \in [\underline{r}_i, \infty)$ because not all locations are equally productive.

Optimal investment in intermittent capacity

What does a distribution of wind capacity costs mean?

Assuming the marginal revenue per K_i is $\tilde{r}_i \geq \underline{r}_i$, then

$$\begin{aligned} K_i &= \overline{K} \Pr(\tilde{r}_i \geq r_i \geq \underline{r}_i) \\ &= \overline{K} \int_{\underline{r}_i}^{\tilde{r}_i} dF(r_i) \\ &= \overline{K} \left(F(\tilde{r}_i) - F(\underline{r}_i)\right) \\ &= \overline{K} F(\tilde{r}_i) \end{aligned}$$

Intuition?

Consumers

There is a continuum of identical consumers of mass one:

- *Instantaneous utility* is $S(q^h)$ where q^h is electricity consumption in state h.[1]
- Instantaneous utility function is strictly concave ($S' > 0$ and $S'' < 0$).
- $S(q^h)$ does not depend on the state.
- von Neumann–Morgenstern utility is

$$\mathbb{E}[S(q)] = \nu S(q^w) + (1 - \nu) S(q^{\overline{w}})$$

Consumers maximize expected utility and are risk-averse (S concave).

[1] By instantaneous utility function we have not specified the complete preference ordering over all states of nature.

Demand

Q: How do we solve the expected utility maximization problem?

Demand

Q: How do we solve the expected utility maximization problem?

- It is a two-stage problem and we solve it using backward induction.
- The instantaneous utility is maximized in each state.

Solving the utility maximization problem $\max_{q^h} S(q^h) - p^h q^h$ implies:

- Inverse demand per consumer is $P(q^h) = S'(q^h)$.
- Direct demand per consumer is $D(p^h) = S'^{-1}(p^h)$.

Emission and environmental policy

- The environmental damage per unit of thermal generation (q_f^h) is $\delta > 0$.
- Emissions are taxed at the Pigouvian rate, i.e. $\tau = \delta$.

When electricity from fossil energy is the only production source (state $\overline{w}$), we assume positive electricity generation is socially efficient, i.e.

$$S'(0) > \underbrace{c + r_f + \delta}_{\substack{\text{Marginal} \\ \text{social costs}}}. \tag{1}$$

Discussions

Does the first fundamental welfare theorem hold, so that a decentralized equilibrium coincide with the solution to the social planner's problem? Why?

The economics of real-time electricity pricing

Optimal energy mix

Simplifying the social planner problem

Before solving the social planner problem to determine the optimal energy mix, we can directly state the solutions to $q_i^{\overline{w}}$, $q_f^{\overline{w}}$, q_i^w, and K_i using intuition:

- $q_i^{\overline{w}} = 0$
- $q_f^{\overline{w}} = K_f$ since thermal capacity is costly (no surplus capacity).
- $q_i^w = K_i$ since intermittent capacity is costly.
- $K_i = \overline{K} F(\tilde{r}_i)$ where $\tilde{r}_i \geq \underline{r}_i$ denotes the marginal revenue per K_i, the value of one more MW of wind, $\nu\gamma_w$ when wind is built (equivalently: the unit costs of the last installed wind turbine), to be determined.

The social planner problem: Welfare function

Social planner then chooses $K_f, \tilde{r}_i, q_f^w, q_r^w, q_r^{\overline{w}}$ and $q_{\overline{r}}$ so as to maximize expected social welfare:

$$\begin{aligned}\mathbb{E}[W] =& \beta\left[\nu S(q_r^w) + (1-\nu)S(q_r^{\overline{w}})\right] + (1-\beta)S(q_{\overline{r}}) \\ &- \left(\nu(c+\delta)q_f^w + (1-\nu)(c+\delta)K_f\right) \\ &- \left(\overline{K}\int_{\underline{r}_i}^{\tilde{r}_i} r_i dF(r_i) + r_f K_f\right)\end{aligned} \tag{2}$$

The social planner problem: Constraints

The constraints of the social planner problem are:

$$K_f = \beta q_r^{\overline{w}} + (1-\beta) q_{\overline{r}} \tag{3}$$

$$\overline{K} F(\tilde{r}_i) + q_f^w = \beta q_r^w + (1-\beta) q_{\overline{r}} \tag{4}$$

$$q_f^w \geq 0 \tag{5}$$

$$q_f^w \leq K_f \tag{6}$$

$$\tilde{r}_i \geq \underline{r}_i \tag{7}$$

- Quantity (equilibrium) constraint in state $h = \overline{w}$.
- Quantity (equilibrium) constraint in state $h = w$.
- Domain constraints on thermal generation.
- Break-even capacity profits bounded by lowest capacity costs $\underline{r}_i$.

The Lagrangian

$$
\begin{aligned}
\mathcal{L} =& \beta\left[\nu S(q_r^w) + (1-\nu)S(q_r^{\overline{w}})\right] + (1-\beta)S(q_{\overline{r}}) \\
&- \left(\nu(c+\delta)q_f^w + (1-\nu)(c+\delta)K_f\right) - \left(\overline{K}\int_{\underline{r}_i}^{\tilde{r}_i} r_i dF(r_i) + r_f K_f\right) \\
&+ \nu\gamma_w\left[\overline{K}F(\tilde{r}_i) + q_f^w - \beta q_r^w - (1-\beta)q_{\overline{r}}\right] \\
&+ (1-\nu)\gamma_{\overline{w}}\left[K_f - \beta q_r^{\overline{w}} - (1-\beta)q_{\overline{r}}\right] \\
&+ \nu\left[\underline{\mu}_f q_f^w + \overline{\mu}_f\left(K_f - q_f^w\right) + \underline{\mu}_i\left(\tilde{r}_i - \underline{r}_i\right)\right],
\end{aligned}
$$

where γ_w and $\gamma_{\overline{w}}$ are the shadow prices on the equilibrium constraints in state w and $\overline{w}$, respectively, and $\underline{\mu}_f$, $\overline{\mu}_f$, and $\underline{\mu}_i$ are shadow prices on the domain constraints.

Find and interpret the first-order conditions

1. Group C: the conditions for q_r^w, $q_r^{\overline{w}}$ and $q_{\overline{r}}$, and what each says.
2. Group P: the conditions for q_f^w and K_f, what each says, and which state pays for a MW of thermal capacity.

Stop point

Group C: three conditions and what each says. Group P: two conditions and what each says.

The full questions, the hint and the stretch are on the worksheet.

Solutions 1, Q1–Q2: the questions

Q1. Differentiate $\mathcal{L}$ and simplify. Group C: q_r^w, $q_r^{\overline{w}}$, $q_{\overline{r}}$. Group P: q_f^w, K_f.

Q2. What does each condition say? Group P: which state pays for a MW of thermal capacity?

The first-order conditions

$$S'(q_r^w) - \gamma_w = 0 \tag{12}$$

$$S'(q_r^{\overline{w}}) - \gamma_{\overline{w}} = 0 \tag{13}$$

$$S'(q_{\overline{r}}) - \nu\gamma_w - (1-\nu)\gamma_{\overline{w}} = 0 \tag{14}$$

$$-(c+\delta) + \gamma_w - \overline{\mu}_f + \underline{\mu}_f = 0 \tag{15}$$

$$(1-\nu)(\gamma_{\overline{w}} - c - \delta) - r_f + \nu\overline{\mu}_f = 0 \tag{16}$$

$$-\tilde{r}_i + \nu\gamma_w + \nu\underline{\mu}' = 0, \qquad \underline{\mu}' \equiv \frac{\underline{\mu}_i}{\overline{K} f(\tilde{r}_i)} \tag{17}$$

From $\partial\mathcal{L}/\partial x = 0$ for $x = q_r^w, q_r^{\overline{w}}, q_{\overline{r}}, q_f^w, K_f, \tilde{r}_i$. What each condition says is part 1 of the worksheet.

First-order conditions II

Isolating for γ_w, $\gamma_{\overline{w}}$ and combining equations yield:

$$S'(q_r^w) = c + \delta + \overline{\mu}_f - \underline{\mu}_f \tag{18}$$

$$S'(q_r^{\overline{w}}) = c + \delta + \frac{r_f - \nu\overline{\mu}_f}{1-\nu} \tag{19}$$

$$S'(q_{\overline{r}}) = \nu S'(q_r^w) + (1-\nu)S'(q_r^{\overline{w}}) \tag{20}$$

$$S'(q_r^w) = \frac{\tilde{r}_i}{\nu} - \underline{\mu}_i' \tag{21}$$

- Defines optimal solution on wholesale market in $h = \overline{w}$.
- Defines optimal solution in retail market.
- RHS in (18): marginal "costs" of thermal generation on wholesale market in $h = w$.
- RHS in (21): marginal "costs" of intermittent generation on wholesale market in $h = w$.

First-order conditions II

Isolating for γ_w, $\gamma_{\overline{w}}$ and combining equations yield:

$$S'(q_r^w) = c + \delta + \overline{\mu}_f - \underline{\mu}_f \tag{18}$$

$$S'(q_r^{\overline{w}}) = c + \delta + \frac{r_f - \nu\overline{\mu}_f}{1-\nu} \tag{19}$$

$$S'(q_{\overline{r}}) = \nu S'(q_r^w) + (1-\nu)S'(q_r^{\overline{w}}) \tag{20}$$

$$S'(q_r^w) = \frac{\tilde{r}_i}{\nu} - \underline{\mu}_i' \tag{21}$$

Combining (18) and (21) provides the condition for the wholesale market in $h = w$:

$$\frac{\tilde{r}_i}{\nu} - \underline{\mu}_i' = c + \delta + \overline{\mu}_f - \underline{\mu}_f \tag{22}$$

Characterizing the optimal energy mix

The solution to the welfare maximization depends on parameter values, because they give rise to different corner solutions.

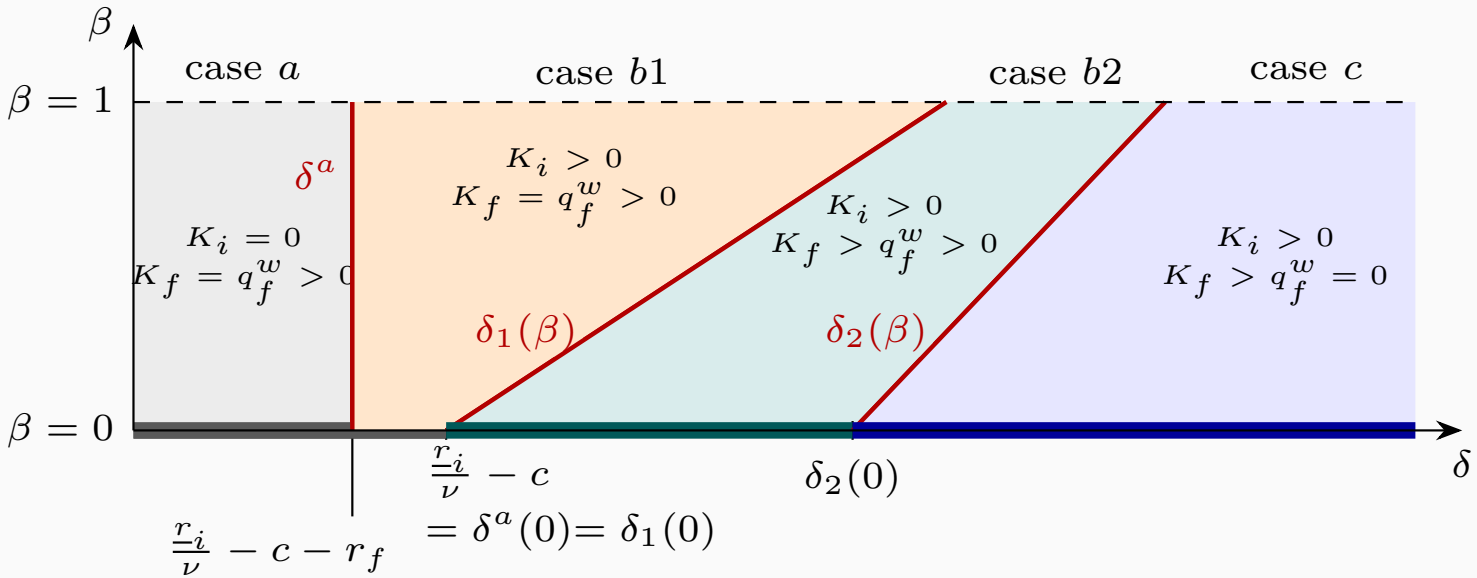
Let us consider two critical parameters: β and δ

- A higher value of β implies more consumer demand is allocated toward $h = w$, with less need for thermal capacity in state $h = \overline{w}$.
- A higher value of δ increases the social costs of using fossil generation, so intermittent generation becomes relatively cheaper.

We go through the cases one by one as δ rises: which constraint binds, what the prices are, and which threshold ends the case.

Graphical illustration of the optimal energy mix

Optimal energy mix with a share of reactive consumers.



Source: Redrawn from Fig. 1 in Ambec and Crampes (2021).

Threshold between b1 and b2: $\delta_1(\beta)$

In case b1, $q_f^w = K_f$. Subtracting (3) from (4), K_f cancels, and wind output in $h = w$ equals the extra consumption of reactive consumers in $h = w$:

$$\overline{K}F(\tilde{r}_i) = \beta\left(q_r^w - q_r^{\overline{w}}\right).$$

This condition holds throughout case b1. Case b1 ends when $c + \delta$ has caught up with p^w, i.e. $\tilde{r}_i = \nu(c + \delta)$ by (22) with $\underline{\mu}_i = \overline{\mu}_f = \underline{\mu}_f = 0$. Inserting the prices at that point into the condition defines $\delta_1(\beta)$:

$$\beta\left[\underbrace{D(c+\delta_1)}_{q_r^w} - \underbrace{D\left(c+\delta_1+\frac{r_f}{1-\nu}\right)}_{q_r^{\overline{w}}}\right] = \overline{K}F(\nu(c+\delta_1)) \qquad (8)$$

- Left: the reactive consumers' shift at the b2 prices. Right: wind capacity at the b2 price in $h = w$.

Threshold between b2 and c: $\delta_2(\beta)$

Case b2 ends when thermal output in $h = w$ reaches zero, $q_f^w = 0$. Then wind alone covers demand in $h = w$ in (4), still at the b2 prices $p^w = c + \delta$, $p = c + \delta + r_f$ and $\tilde{r}_i = \nu(c + \delta)$:

$$\beta \underbrace{D(c + \delta_2)}_{q_r^w} + (1 - \beta) \underbrace{D(c + \delta_2 + r_f)}_{q_{\overline{r}}} = \overline{K} F(\nu(c + \delta_2)) \tag{9}$$

- Left: total demand in $h = w$ at the b2 prices. Right: wind capacity at the b2 price in $h = w$.
- Below δ_2, thermal fills the gap in $h = w$; above it, wind covers all demand in $h = w$ (case c).
- $\delta_1(\beta) < \delta_2(\beta)$.

Summary: Proposition 1

Proposition 1: Optimal capacity, output, and decentralizing prices:

Case	Condition	Prices and Quantities
a)	$\delta < \delta^a \equiv \frac{r_i}{\nu} - (c + r_f)$ No intermittent investment	$p = p^w = p^{\overline{w}} = c + \delta + r_f$ $K_i = 0,\ K_f = q_f^w = D(c + \delta + r_f)$
b1)	$\delta^a \leq \delta \leq \delta_1(\beta)$	$p^w = \frac{\tilde{r}_i}{\nu},\ p^{\overline{w}} = \frac{c+\delta+r_f-\tilde{r}_i}{1-\nu}$ †, $p = c + r_f + \delta$ $K_i = \overline{K}F(\nu p^w),\ K_f = \beta D(p^{\overline{w}}) + (1-\beta)D(p),\ q_f^w = K_f$ $\tilde{r}_i$: $\overline{K}F(\tilde{r}_i) = \beta\left[D\left(\frac{\tilde{r}_i}{\nu}\right) - D\left(\frac{c+\delta+r_f-\tilde{r}_i}{1-\nu}\right)\right]$
b2)	$\delta_1(\beta) \leq \delta \leq \delta_2(\beta)$	$p^w = c + \delta = \frac{\tilde{r}_i}{\nu},\ p^{\overline{w}} = c + \delta + \frac{r_f}{1-\nu},\ p = c + r_f + \delta$ $K_i = \overline{K}F(\nu p^w),\ K_f = \beta D(p^{\overline{w}}) + (1-\beta)D(p),\ q_f^w = K_f - K_i + \beta\left(q_r^w - q_r^{\overline{w}}\right) > 0$†
c)	$\delta > \delta_2(\beta)$ Only intermittent in w	$p^w = \frac{\tilde{r}_i}{\nu},\ p^{\overline{w}} = c + \delta + \frac{r_f}{1-\nu},\ p = \tilde{r}_i + (1-\nu)(c + \delta) + r_f$ $K_i = \overline{K}F(\nu p^w),\ K_f = \beta D(p^{\overline{w}}) + (1-\beta)D(p),\ q_f^w = 0$ $\tilde{r}_i$: $\overline{K}F(\tilde{r}_i) = \beta D\left(\frac{\tilde{r}_i}{\nu}\right) + (1-\beta)D\left((1-\nu)(c+\delta) + r_f + \tilde{r}_i\right)$

In case b) both sources are used in state w. †Corrects a typo in Proposition 1 of Ambec and Crampes (2021); see their eqs. (25) and (27).

Why not just build more wind instead of thermal capacity?

In cases b1, b2 and c, wind is cheap enough to be built. Yet thermal capacity is still needed in every case:

$$K_f = \beta D(p^{\overline{w}}) + (1 - \beta)D(p).$$

Q: Why not just build more wind instead of thermal capacity?

Intuition?

The economics of real-time electricity pricing

Consumption and carbon emissions

Emissions and RTP

The effect of RTP on emissions is determined by the sign of:

$$\frac{d\mathbb{E}[q_f]}{d\beta} = \frac{d}{d\beta}\left(\nu q_f^w + (1-\nu)K_f\right)$$

Corollary 1: When energy supply is intermittent, increasing the share of reactive consumers reduces emissions in case *c*. In case *b2*, it reduces emissions only if demand is strictly concave (no effect if demand is linear).† It also reduces emissions in case *b1* as long as β is not too large.

† Differs from Corollary 1 in Ambec and Crampes (2021). With q_f^w from Proposition 1, K_i does not depend on β in case b2, so thermal generation absorbs the whole change in consumption.

Emissions and RTP: who emits more?

For a given energy mix, a reactive consumer emits more than a non-reactive one if and only if

$$\nu \underbrace{\frac{q_f^w}{K_i + q_f^w}}_{\text{thermal share in } w} (q_r^w - q_{\overline{r}}) > (1-\nu) \underbrace{1}_{\text{in } \overline{w}} \left(q_{\overline{r}} - q_r^{\overline{w}}\right).$$

The thermal share is the relative emission factor: the emissions per MWh in $h = w$ relative to $h = \overline{w}$, where all power is thermal.

- Left: expected extra emissions in $h = w$, the thermal share times the extra consumption $q_r^w - q_{\overline{r}}$.
- Right: expected emissions saved in $h = \overline{w}$.
- **Case c:** $q_f^w = 0$, so the left side is zero. Shifting consumption to $h = w$ adds no emissions, and reactive consumers always lower them.

Emissions and RTP: cases b2 and b1

- **Case b2:** thermal is marginal in both states and K_i is fixed, so emissions follow average consumption:

$$\frac{d\mathbb{E}[q_f]}{d\beta} = \nu D(p^w) + (1-\nu)D(p^{\overline{w}}) - D(p).$$

  Zero with linear demand; negative with strictly concave demand (Jensen).

- **Case b1:** thermal runs at full capacity in both states, so expected thermal output is $K_f = \beta D(p^{\overline{w}}) + (1-\beta)D(p)$, with p fixed and $p^{\overline{w}} = \frac{c+\delta+r_f-\tilde{r}_i}{1-\nu}$:

$$\frac{dK_f}{d\beta} = \underbrace{D(p^{\overline{w}}) - D(p)}_{<0} + \underbrace{\beta D'(p^{\overline{w}})\frac{dp^{\overline{w}}}{d\beta}}_{>0}, \qquad \frac{dp^{\overline{w}}}{d\beta} = -\frac{1}{1-\nu}\frac{d\tilde{r}_i}{d\beta} < 0.$$

  One more reactive consumer uses less in $h = \overline{w}$, but more wind lowers $p^{\overline{w}}$ and all reactive consumers use more there. The second term vanishes as $\beta \to 0$.

Source: Ambec and Crampes (2021), p. 5; $d\tilde{r}_i/d\beta > 0$ by their eq. (36).

The economics of real-time electricity pricing

The social benefit of reactive consumers

Welfare effect of RTP

The optimum $X = (K_f, \tilde{r}_i, q_f^w, q_r^w, q_r^{\overline{w}}, q_{\overline{r}})'$ depends on β. At the optimum $\mathbb{E}[W] = \mathcal{L}$, so by the chain rule:

$$\frac{d\mathbb{E}[W]}{d\beta} = \underbrace{\frac{\partial \mathcal{L}}{\partial X}}_{=0} \frac{\partial X}{\partial \beta} + \frac{\partial \mathcal{L}}{\partial \beta}$$

β enters $\mathcal{L}$ only in $\beta \left[\nu S(q_r^w) + (1-\nu) S(q_r^{\overline{w}})\right] + (1-\beta) S(q_{\overline{r}})$ and in market clearing via $-\beta q_r^h - (1-\beta) q_{\overline{r}}$, weighted by $\nu\gamma_w$ and $(1-\nu)\gamma_{\overline{w}}$. With $p^h = \gamma_h$:

$$\begin{aligned}\frac{d\mathbb{E}[W]}{d\beta} = &\left[\nu S(q_r^w) + (1-\nu) S(q_r^{\overline{w}})\right] - S(q_{\overline{r}}) \\ &- \nu p^w \left(q_r^w - q_{\overline{r}}\right) - (1-\nu) p^{\overline{w}} \left(q_r^{\overline{w}} - q_{\overline{r}}\right) \qquad (11)\end{aligned}$$

(11) holds in all cases of Proposition 1.

Welfare effect of RTP in cases b2 and c

Via rewritten $\mathbb{E}[W]$

In cases b2 and c, $\nu p^w = \tilde{r}_i$ and $(1-\nu)p^{\overline{w}} = (1-\nu)(c+\delta) + r_f$, so (11) becomes[†]

$$\frac{d\mathbb{E}[W]}{d\beta} = \left[\nu S(q_r^w) + (1-\nu)S(q_r^{\overline{w}})\right] - S(q_{\overline{r}}) - \left[(1-\nu)(c+\delta) + r_f\right]\left(q_r^{\overline{w}} - q_{\overline{r}}\right) - \tilde{r}_i\left(q_r^w - q_{\overline{r}}\right) \tag{10}$$

(10) is (11) with prices replaced by costs: costs saved on thermal energy and capacity in $h = \overline{w}$, minus the cost of serving the extra consumption in $h = w$. This is possible in b2 and c because thermal capacity does not bind in $h = w$, so the prices contain no scarcity rent and are given by the cost parameters alone.

[†] Ambec and Crampes (2021) state (10) for all cases. In case b1, the thermal capacity constraint binds in $h = w$, and (10) omits a term. Their eq. (11) holds in all cases.

Consumer welfare effect of RTP

Lemma 2: If demand is not too convex ($S(D(p))$ concave in p),[†] the expected gross surplus of being a reactive consumer is smaller than the gross surplus of being non-reactive:

$$S(q_{\overline{r}}) > \nu S(q_r^w) + (1-\nu) S(q_r^{\overline{w}})$$

[†] Ambec and Crampes (2021) state Lemma 2 without this condition. It is their condition (34); see their footnote 21: concavity of S is not sufficient. With constant-elasticity demand, the inequality is reversed.

Consumer welfare effect of RTP

Lemma 2: If demand is not too convex ($S(D(p))$ concave in p), the expected gross surplus of being a reactive consumer is smaller than the gross surplus of being non-reactive:

$$S(q_{\overline{r}}) > \nu S(q_r^w) + (1-\nu) S(q_r^{\overline{w}})$$

Proof: a reactive consumer buys $D(p^h)$ in each state, a non-reactive one buys $D(p)$ at the average price $p = \nu p^w + (1-\nu)p^{\overline{w}}$. If $S(D(p))$ is concave in p, Jensen's inequality gives

$$\nu S(D(p^w)) + (1-\nu) S(D(p^{\overline{w}})) \leq S(D(p)) = S(q_{\overline{r}}),$$

strictly when $p^w \neq p^{\overline{w}}$ and $S(D(p))$ is strictly concave. **Intuition?**

Total welfare effect of RTP

Proposition 2: When the energy mix contains intermittent sources, the impact of an increase in demand response on welfare is threefold:

(i) a loss from exposing risk-averse consumers to volatile consumptions (if demand is not too convex, Lemma 2),

(ii) lower costs and emissions from reducing thermal powered electricity,[†]

(iii) increased costs from the installation of more renewables.[†]

The net impact is positive (i.e. $d\mathbb{E}[W]/d\beta > 0$).

[†] In case b2, K_i does not depend on β: no new renewables are installed, and emissions fall only if demand is strictly concave (Corollary 1). Ambec and Crampes (2021) state (ii) and (iii) without this qualification.

Total welfare effect of RTP: the three effects in (10)

In cases b2 and c, the three effects of Proposition 2 are the three parts of (10):

$$\frac{d\mathbb{E}[W]}{d\beta} = \underbrace{\left[\nu S(q_r^w) + (1-\nu)S(q_r^{\overline{w}})\right] - S(q_{\overline{r}})}_{\text{(i): loss from volatile consumption, Lemma 2}}$$

$$\underbrace{-\left[(1-\nu)(c+\delta) + r_f\right]\left(q_r^{\overline{w}} - q_{\overline{r}}\right)}_{\text{(ii): thermal costs and emissions saved in } h=\overline{w}}$$

$$\underbrace{-\tilde{r}_i\left(q_r^w - q_{\overline{r}}\right)}_{\text{(iii): cost of serving more consumption in } h=w}$$

- δ is the price of emissions, so the saved emissions are inside $c+\delta$ in (ii).

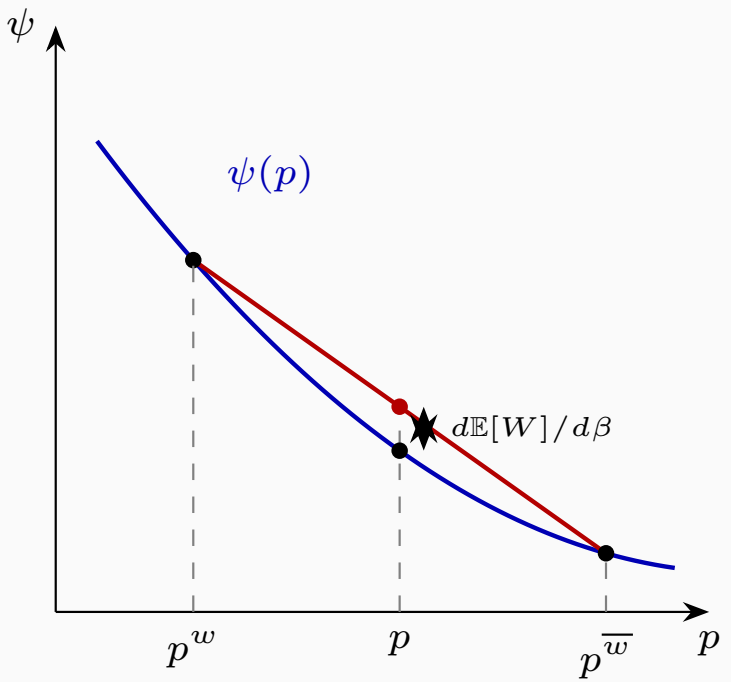
Why the net impact is positive

Net surplus: $\psi(p) \equiv \max_q S(q) - pq = S(D(p)) - pD(p)$.

Multiply out (11) and use $p = \nu p^w + (1-\nu)p^{\overline{w}}$:

$$\frac{d\mathbb{E}[W]}{d\beta} = \nu\psi(p^w) + (1-\nu)\psi(p^{\overline{w}}) - \psi(p)$$

- ψ is convex: a maximum of functions linear in p (or $\psi'' = -D' > 0$).
- Jensen: $\mathbb{E}[\psi(p^h)] \geq \psi(\mathbb{E}[p^h]) = \psi(p)$, so $d\mathbb{E}[W]/d\beta \geq 0$.



The chord lies above the convex curve.

Source: Ambec and Crampes (2021), eq. (11) and Appendix B.3.

How much is one more reactive consumer worth?

By the previous slide, $\dfrac{d\mathbb{E}[W]}{d\beta} = \mathbb{E}[\psi(p^h)] - \psi(p)$, with $\psi(p) = S(D(p)) - pD(p)$.

Derivatives, using $S'(D(p)) = p$:

$$\psi'(p) = S'(D(p))D'(p) - D(p) - pD'(p) = -D(p),$$
$$\psi''(p) = -D'(p) > 0.$$

Second-order Taylor expansion of $\psi(p^h)$ around p:

$$\psi(p^h) \approx \psi(p) - D(p)(p^h - p) - \tfrac{1}{2}D'(p)(p^h - p)^2$$

Taking expectations:

$$\mathbb{E}[\psi(p^h)] \approx \psi(p) - D(p)\,\mathbb{E}[p^h - p] - \tfrac{1}{2}D'(p)\,\mathbb{E}\left[(p^h - p)^2\right]$$

How much is one more reactive consumer worth?

$$\mathbb{E}[\psi(p^h)] \approx \psi(p) - D(p)\,\mathbb{E}[p^h - p] - \tfrac{1}{2} D'(p)\,\mathbb{E}\left[(p^h - p)^2\right]$$

- $\mathbb{E}[p^h - p] = 0$: p is defined as the expected price, $p = \nu p^w + (1-\nu)p^{\overline{w}}$ (the fair retail price, (14)).
- Variance: with $\Delta \equiv p^w - p^{\overline{w}}$, $p^w - p = (1-\nu)\Delta$ and $p^{\overline{w}} - p = -\nu\Delta$, so

$$\mathrm{Var}(p^h) = \nu(1-\nu)^2\Delta^2 + (1-\nu)\nu^2\Delta^2 = \nu(1-\nu)\Delta^2.$$

Hence

$$\frac{d\mathbb{E}[W]}{d\beta} \approx \tfrac{1}{2}\,|D'(p)|\,\mathrm{Var}(p^h), \qquad \mathrm{Var}(p^h) = \nu(1-\nu)\left(p^w - p^{\overline{w}}\right)^2$$

Intuition?

Diminishing returns to RTP

Proposition 3: The expected marginal social welfare is non-increasing with the proportion of reactive consumers β,[†] i.e.

$$\frac{d^2\mathbb{E}[W]}{d\beta^2} \leq 0, \quad \text{strictly in cases b1 and c.}$$

- Why: by the Taylor approximation (ours), $\frac{d^2\mathbb{E}[W]}{d\beta^2} \approx \frac{1}{2}|D'(p)|\frac{d\,\mathrm{Var}(p^h)}{d\beta}$.
  In cases b1 and c, more reactive consumers move p^w and $p^{\overline{w}}$ closer, so Var falls: negative.
- In case b2, thermal plants set the price in both states, so Var and the marginal gain do not depend on β: zero.
- If raising β is associated with investment costs (e.g. for smart-meters) and the gain is strictly decreasing, then it might not be optimal to let all consumers be subject to RTP.

[†] Ambec and Crampes (2021) state "decreasing". Their proof (Appendix C) gives $d^2\mathbb{E}[W]/d\beta^2 = 0$ in case b2.

Why exposing consumers to prices still pays

One consumer with $S(q) = 200q - 5q^2$, so $D(p) = 20 - 0.1p$. Two equally likely states ($\nu = \frac{1}{2}$) with $p^w = 40$ and $p^{\overline{w}} = 120$. The fixed retail price is the fair price $p = 80$.

1. Lemma 2: the three quantities and the expected gross surplus of each consumer.
2. Proposition 2: the two price terms in (11), $d\mathbb{E}[W]/d\beta$, and the check against $\frac{1}{2}|D'(p)|\,\mathrm{Var}(p^h)$.
3. Proposition 3: the same at prices 60 and 100.

Stop point

Three quantities, two expected gross surpluses and their difference, the sum of the two price terms in (11), and $d\mathbb{E}[W]/d\beta$ at both price pairs.

The full questions and the stretch are on the worksheet.

Solutions 2, Q1–Q3: the questions

Q1. Lemma 2: find q_r^w, $q_r^{\overline{w}}$ and $q_{\overline{r}}$. Who has the larger expected gross surplus?

Q2. Proposition 2: the two price terms in (11) and their sum; $d\mathbb{E}[W]/d\beta$ against $\frac{1}{2}|D'(p)|\,\mathrm{Var}(p^h)$.

Q3. Proposition 3: what is the next reactive consumer worth at prices 60 and 100?

Does real-time pricing provide more flexibility?

A quasi-natural experiment

Ambec and Crampes (2021) assume that consumers on RTP respond to prices. We now ask whether this is true: does RTP make $|D'(p)|$ large?

In Spain, RTP became the default contract in 2015, rolled out gradually.

Fabra et al. (2021) use this variation to estimate the price elasticity of demand for the two consumer groups.

The design of Fabra et al. (2021)

Fabra et al. (2021): since October 2015, RTP is the default tariff for Spanish households.

- **Tariff:** time-invariant network charge; the energy charge (about half the price) follows the day-ahead price hour by hour, published at 20.30 the day before. Households can choose a time-of-use network tariff and/or opt out of RTP with a commercial retailer.
- **Data:** hourly consumption from two utilities, Jan. 2016–June 2017, random sample of 21,233 households, 48.2% on RTP. The smart meter rollout followed a national plan (plausibly exogenous).
- **Identification:** the price is endogenous (high when demand is high). Instrument z_t: the day-ahead forecast of national wind production.

The design of Fabra et al. (2021): estimation

Estimated household by household with 2SLS:

$$y_{it} = \beta_{i0} + \beta_{i1}\hat{p}_t + \Omega_i X_t + \lambda_i W_{it} + u_{it}$$

- y_{it}: consumption of household i in hour t. $\hat{p}_t$: the price, instrumented with the wind forecast z_t.
- X_t: hour of day, month, Fourier terms, system demand. W_{it}: household controls (e.g. temperature bins).
- β_{i1}: household i's price response.

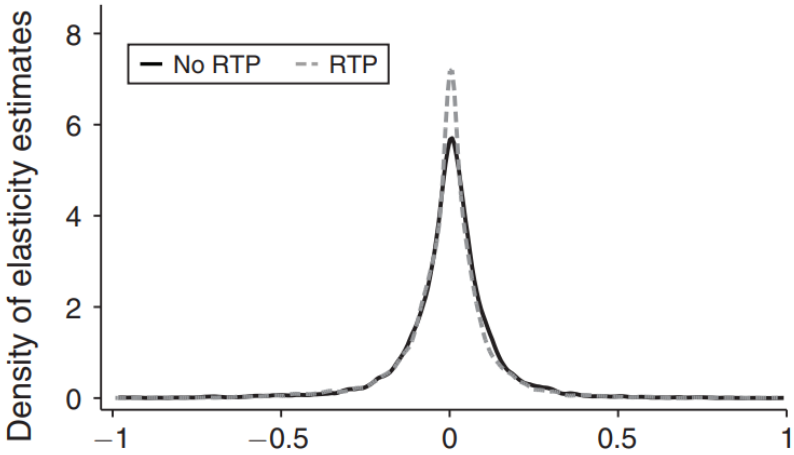
The price varied little: the ratio of the day's highest to lowest price was 1.23 on average.

Evidence from Spain: no more elastic demand

Fabra et al. (2021): No, the rollout of RTP in Spain did *not* provide a more elastic electricity demand:

FIGURE 1. RTP VERSUS NO-RTP ELASTICITIES

Panel A. Density for specification (2)



Panel B. Mean estimates with alternative specifications

	(1)	(2)	(3)	(4)
RTP	−0.054	−0.0072	−0.014	−0.017
	(0.0025)	(0.0019)	(0.0016)	(0.0027)
No RTP	−0.058	−0.0031	−0.011	−0.013
	(0.0028)	(0.0022)	(0.0016)	(0.0030)

Notes: Table shows mean elasticities by type of tariff (RTP versus non-RTP). Standard errors clustered at the postal code level. $N = 17,928$. Individual elasticity estimates using (1) block and temperature bin fixed effects and interactions, plus block times solar output; (2) all variables in (1) plus aggregate demand as a control; (3) all in (1) and (2) but with temperature and temperature squared instead of temp bins; and (4) post-lasso with Fourier transforms at daily, weekly, and annual frequency interacted with aggregate demand, solar production, temperature, and temperature square.

Source: Fabra et al. (2021, p. 428, Figure 1)

Why is demand inelastic? I

Two possible explanations for inelastic demand:

1. Consumers are rationally price inelastic: They are fully informed about the costs of using appliances and

$$\text{marginal costs} > \text{marginal benefits}$$

2. Consumers are not fully informed about price and quantity, or they do not have sufficient capacity to change behaviour.

Explanation 1 is inside the model: a small $|D'(p)|$ gives a small gain. Explanation 2 is outside the model, which assumes that reactive consumers know and respond to p^h.

Only in the latter case should we do something about it.

Why is demand inelastic? II

Evidence for **explanation 1**: The benefit of changing your consumption was at most 5% of an average electricity bill.

Evidence for **explanation 2**:

- Weak price signal:
  - 77% of consumers did not know what the difference between the two contracts was.
  - 64% of consumers did not know which contract they had.
  - Only 9% of the customers of the one utility that offered a price app used it.
- Unknown product attributes: Consumers know the price but not the quantity (Jessoe and Rapson, 2014).

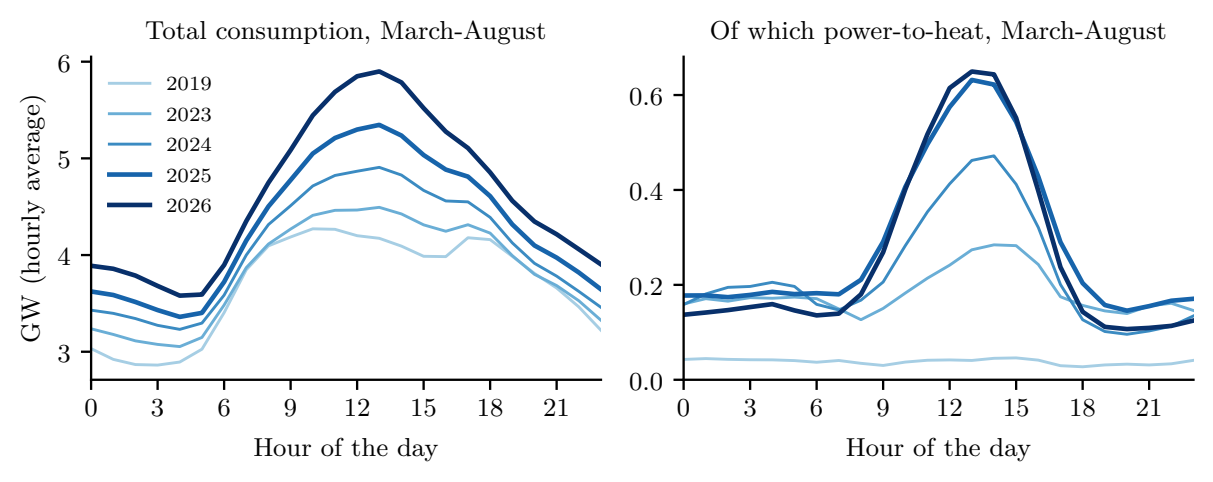
Elements of flexible demand

Two sufficient conditions for demand to become flexible seem to be (Harding and Lamarche, 2016; Blonz et al., 2021):

- RTP: dynamic price incentives.
- Smart technologies and "load switching" devices: enable reallocation of consumption at low marginal cost.

Back to the model: smart meters are the cost of raising β (Proposition 3); load-switching devices raise $|D'(p)|$.

Something is happening: the midday bump



Source: Own calculations based on `www.energidataservice.dk`. DK1 and DK2, March–August.

- About half of the growth in the midday bump since 2023 is power-to-heat. This is consistent with automated demand moving to cheap solar hours, but we have not shown that it responds to prices.
- About a third of Danish households respond significantly to the hourly price (Andersen and Dietrich, 2026).

Q: Automated devices respond to the same spot price at the same time. What does that do to the grid?

Relieving the grid: flexible demand and network pricing

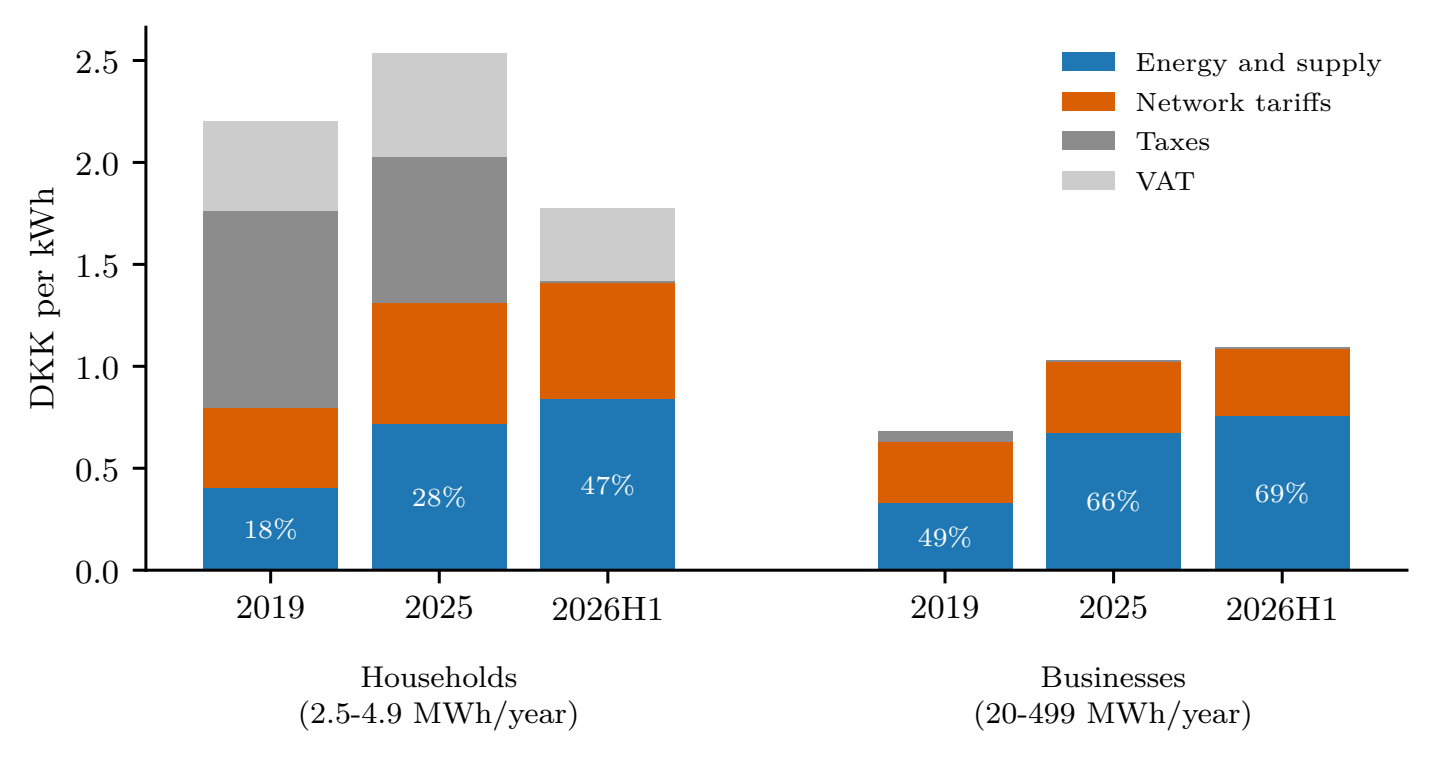
From energy scarcity to grid scarcity

Recall: flexible demand has two jobs.

- So far: energy scarcity. Signal: the spot price.
- Now: grid scarcity. Signal: the network tariff.

The policy focus has moved to the grid. But has the price focus switched to network tariffs?

What does a Danish consumer pay?



Source: Statistics Denmark (ENERGI1, ENERGI2). Businesses excl. VAT and refundable taxes. Energy and supply includes the supplier's margin.

Until 2025, energy was about a quarter of the household price. After the cut in the electricity tax in 2026, it is about half.

The role of power grids

Power grids are the backbone of electrification:

- They connect supply and demand across locations. The grid is a common-pool resource subject to congestion externalities: its capacity is shared, and one user's consumption when the grid is full leaves less for the others.
- They are a necessary (but not sufficient) condition for electrification.

Capacity adequacy is key for electrification

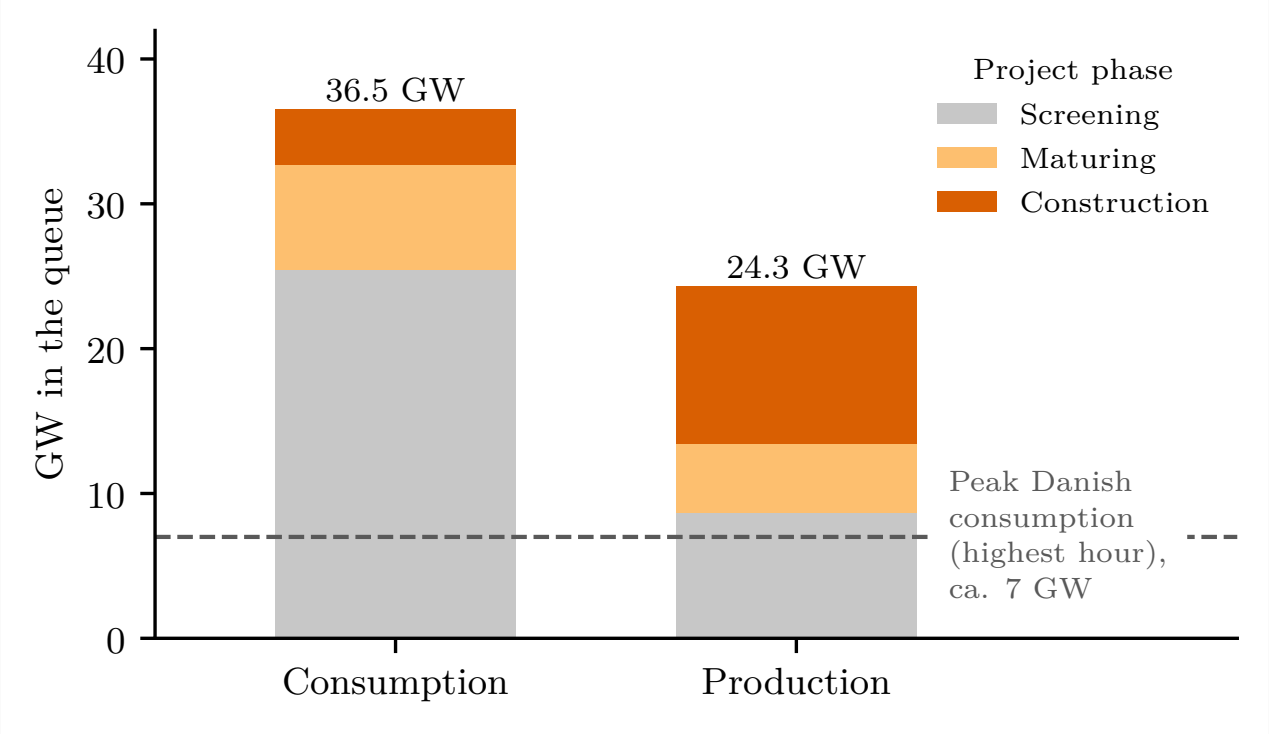
The two main tasks of power grid operators (TSOs/DSOs) are:

1. *Capacity adequacy:* ability of the grid to handle peak supply and load at any given moment.
2. *Congestion management:* sending the right price signals to internalize congestion externalities (bottlenecks).

The two are strategic complements (due to positive feedback loops):

- If capacity is inadequate, bottlenecks will occur and create congestion costs in the system.
- Internalized congestion costs generate revenue for capacity investments.

The queue for the transmission grid is five times peak demand



Source: Energinet, Overblik over nettilslutningsprojekter (data 1 October 2026). Peak consumption: Energinet, 16 June 2026.

Transmission projects are delayed

Energinet (the Danish TSO) manages the transmission network in Denmark. According to the National Audit Office, covering 2023–2025 (Rigsrevisionen, 2026):

- About 70% of Energinet's projects are delayed.
- The average delay is about 2½ years.
- The expansion is more than DKK 10 billion more expensive than planned.

Energinet's own status, Q2 2026 (Energinet, 2026): 43% of projects on time, delayed projects on average 30.6 months late. Energinet's reasons: longer environmental approvals, longer delivery times for components, and an organisation that did not scale fast enough.

Grid capacity will be scarce for years. This is when flexible demand is worth most.

Adding the grid to the model

Our extension (not in Ambec and Crampes (2021)): the grid carries at most G, at cost r_G per unit. Add to the Lagrangian

$$\nu\lambda_w \left[G - \beta q_r^w - (1-\beta)q_{\overline{r}}\right] + (1-\nu)\lambda_{\overline{w}} \left[G - \beta q_r^{\overline{w}} - (1-\beta)q_{\overline{r}}\right] - r_G G.$$

$$S'(q_r^h) = \gamma_h + \lambda_h$$
$$\partial\mathcal{L}/\partial G = \nu\lambda_w + (1-\nu)\lambda_{\overline{w}} - r_G = 0$$

- Efficient price = energy price γ_h + grid price $\lambda_h \geq 0$, positive only when the grid is full. Seeing only the spot price, consumers use too much then.
- Suppose the grid is full only in w, as in the stretch of part 3 (full only in $\overline{w}$ is impossible with $\beta > 0$): $\lambda_w = r_G/\nu$, the same form as (16). Peak-load pricing (Steiner, 1957; Boiteux, 1960).

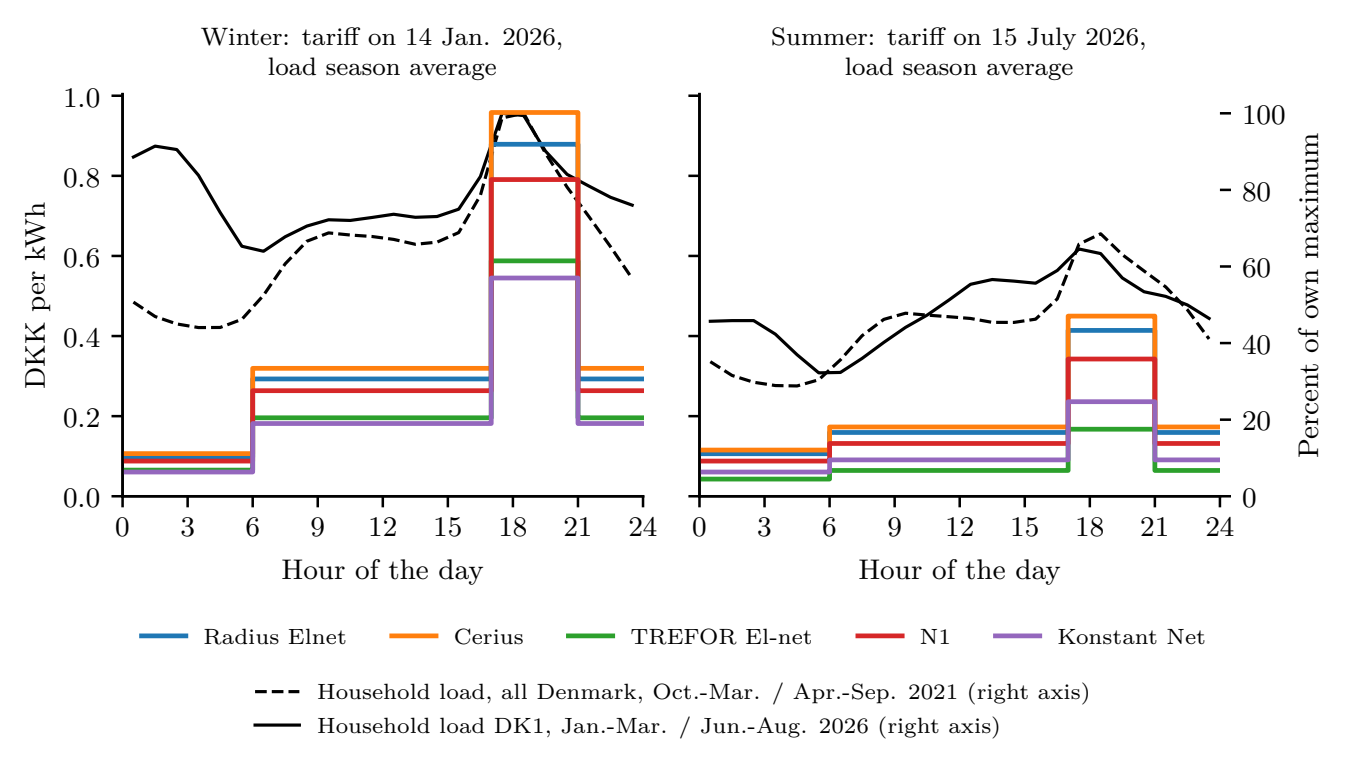
Recall: zonal vs. nodal pricing

The grid scarcity price λ_h differs between locations. Where does Denmark price it?

Scarcity	**Efficient signal**	**Denmark**
Energy	Spot price per hour (quarter)	Spot price per bidding zone (DK1, DK2)
Transmission grid	Nodal price: energy + congestion at each node	Zonal price: congestion within a zone is solved by redispatch, not priced
Distribution grid	Local, dynamic price	Time-of-use tariff, national time windows

A zonal price ignores λ_h within the zone, so there is no signal of *where* the grid is full. Danish law allows geographically differentiated tariffs for producers, but for consumers only at 10 kV and above and for local associations of network users (Elforsyningsloven § 73, stk. 1).

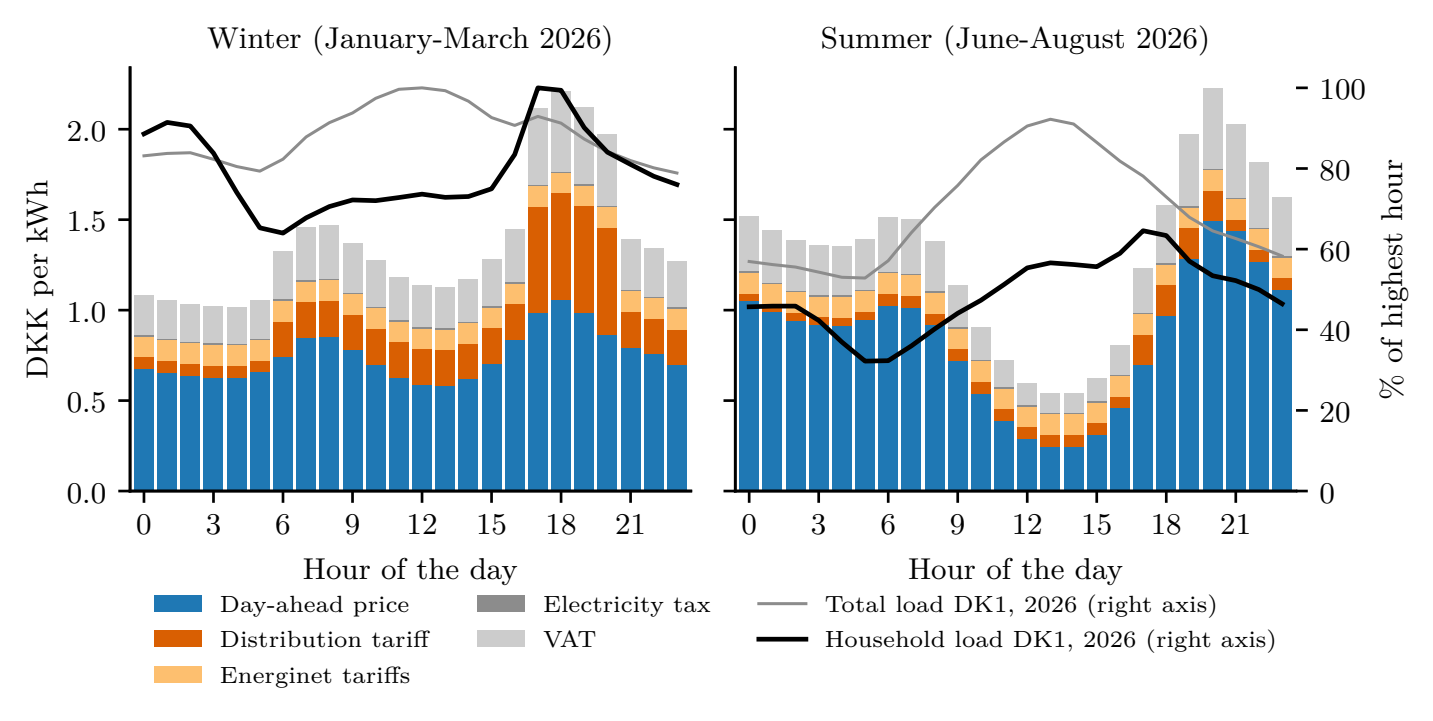
Distribution tariffs: national time-of-use windows



Source: Own calculations based on `www.energidataservice.dk`. Household tariff C. Household load in percent of each series' maximum.

- Same time windows in the whole country (00–06, 06–17 and 21–24, 17–21) and fixed ratios between them. Only the level differs.
- Costs are allocated by the average load in each window, although the argument is peak load (Forsyningstilsynet, 2022; Forsyningstilsynet, 2024).

The hourly price signal of a household



Source: Own calculations based on `www.energidataservice.dk`. Distribution tariff of TREFOR El-net (DK1). Load in DK1, percent of each series' maximum.

Q: Which component signals energy scarcity, and which grid scarcity? Does the tariff peak with household load or with total load?

Grid-friendliness: a second-best answer

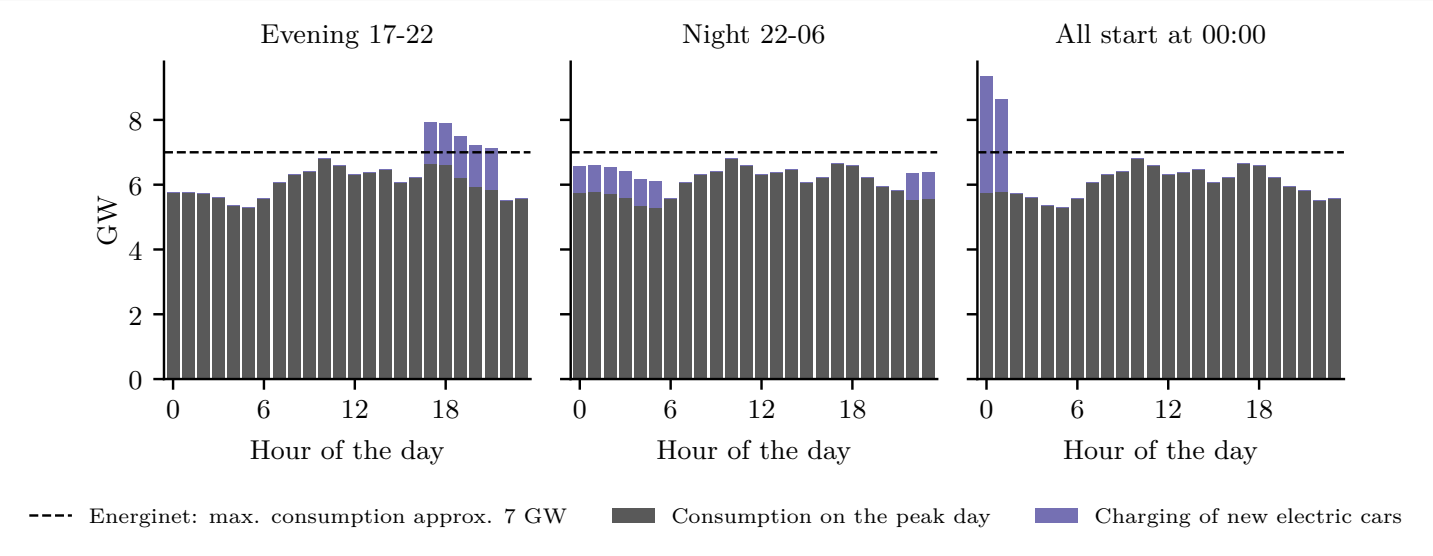
If the price signalled grid scarcity, grid-friendly behaviour would simply be to respond to the price. Because it does not, policy tries to get the behaviour in other ways:

- New projects in the connection queue will be assessed on grid-friendliness: flexibility, location and interaction with other use of the grid. Energinet and the grid companies make the assessment (Klima-, Energi- og Forsyningsministeriet, 2026b).
- Examples: a battery placed where there is spare capacity, or an electric boiler in district heating that can be disconnected at peak load. Data centres are now placed last in the queue (Klima-, Energi- og Forsyningsministeriet, 2026a).

Someone has to decide what is grid-friendly, instead of the price revealing it.

- A rule that everyone follows at the same time can create a new peak.
- Prices vs. quantities: a rule fixes behaviour, a price lets those with the lowest cost of adjusting adjust.

Electric cars and a fixed tariff step



Source: Own calculations. Peak consumption day (excl. self-consumption and grid losses) plus the charging of the about 815,000 electric cars added to the stock by 2030 (KF26). Assumptions: 40% charge on a given night with 11 kW.

"All start at 00:00" is an upper bound on synchronised charging when the low tariff starts, not a forecast.

What is grid-friendly?

1. Propose a rule that defines grid-friendly charging of an electric car.
2. Apply your rule to the electric car scenario. What happens if all cars follow it?
3. Which price signal would make the rule unnecessary? Why is it hard to introduce in Denmark?

Stop point

A fixed rule moves the peak. A price that reflects local grid scarcity spreads it.

The stretch is on the worksheet.

Solutions 3, Q1–Q3: the questions

Q1. Propose a rule that defines grid-friendly charging of an electric car.

Q2. Apply your rule to the scenario. What happens if all cars follow it?

Q3. Which price signal would make the rule unnecessary? Why is it hard to introduce in Denmark?

Key take-aways

- Flexible demand has two jobs: responding to energy scarcity and to grid scarcity.
- More reactive consumers move the energy mix towards renewables and raise welfare, but with diminishing returns (Ambec and Crampes, 2021).
- Real-time prices alone do not make demand flexible. It takes information and automation.
- The spot price signals energy scarcity. Zonal prices and fixed national tariffs do not signal local grid scarcity, so grid-friendliness rules are a second-best answer.

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Appendices

Rewriting the social welfare function I

Go back

Rewrite the social welfare function to

$$\begin{aligned}\mathbb{E}[W] =& \beta\left[\nu S(q_r^w) + (1-\nu)S(q_r^{\overline{w}})\right] + (1-\beta)S(q_{\overline{r}}) \\ &- \left(\nu(c+\delta)q_f^w + (1-\nu)(c+\delta)K_f\right) \\ &- \underbrace{\overline{K}\left[\tilde{r}_i F(\tilde{r}_i) - \underline{r}_i F(\underline{r}_i) - \mathcal{F}(\tilde{r}_i) + \mathcal{F}(\underline{r}_i)\right]}_{\overline{K}\int_{\underline{r}_i}^{\tilde{r}_i} r_i\, dF(r_i)} \\ &- r_f K_f\end{aligned}$$

$$\mathcal{F}(r_i) \equiv \int F(r_i)dr_i$$

Rewriting the social welfare function II

Go back

Substituting the equilibrium definitions in (3) and (4) into $\mathbb{E}[W]$:

$$\begin{aligned}\mathbb{E}[W] =& \beta\left[\nu S(q_r^w) + (1-\nu)S(q_r^{\overline{w}})\right] + (1-\beta)S(q_{\overline{r}}) \\ & - \nu(c+\delta)q_f^w - \left[(1-\nu)(c+\delta) + r_f\right]\left(\beta q_r^{\overline{w}} + (1-\beta)q_{\overline{r}}\right) \\ & - \tilde{r}_i\left[\beta q_r^w + (1-\beta)q_{\overline{r}} - q_f^w\right] + \overline{K}\left[\mathcal{F}(\tilde{r}_i) - \mathcal{F}(\underline{r}_i)\right]\end{aligned}$$

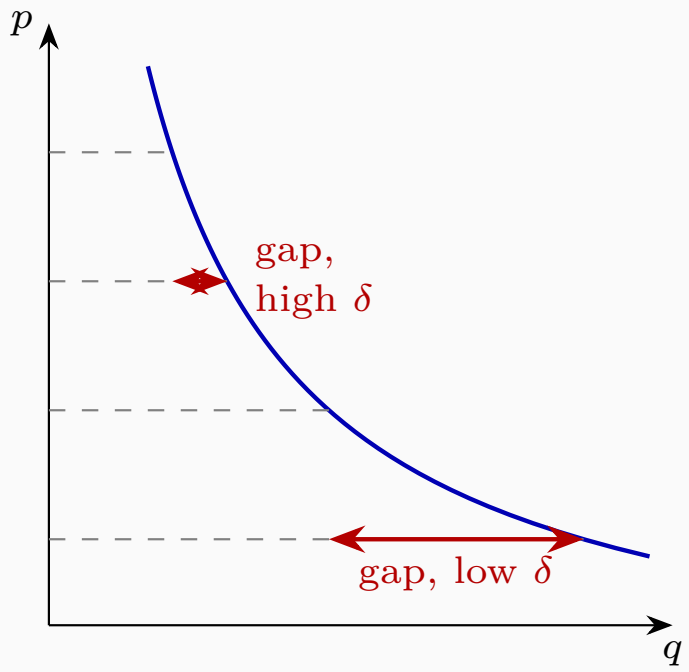
Differentiating this partially with respect to β gives (10). That is valid in cases b2 and c only: in b1 the thermal capacity constraint binds in $h = w$, and its shadow price is missing here.

Why $\delta_1(\beta)$ increases with β

Rearranging eq. (8), the threshold between cases b1 and b2:

$$\beta = \frac{\overline{K} F(\nu(c+\delta_1))}{D(c+\delta_1) - D\left(c+\delta_1+\frac{r_f}{1-\nu}\right)}$$

- Numerator: wind capacity. Increases in δ_1.
- Denominator: consumption gap of reactive consumers between $h = w$ and $h = \overline{w}$. The two prices differ by a fixed $\frac{r_f}{1-\nu}$.
- The gap changes with δ by $D'(c+\delta) - D'\left(c+\delta+\frac{r_f}{1-\nu}\right) \leq 0$ if $D'' \geq 0$.
- So the ratio increases in δ_1, and $\delta_1(\beta)$ increases in β. Without convexity the sign can flip.

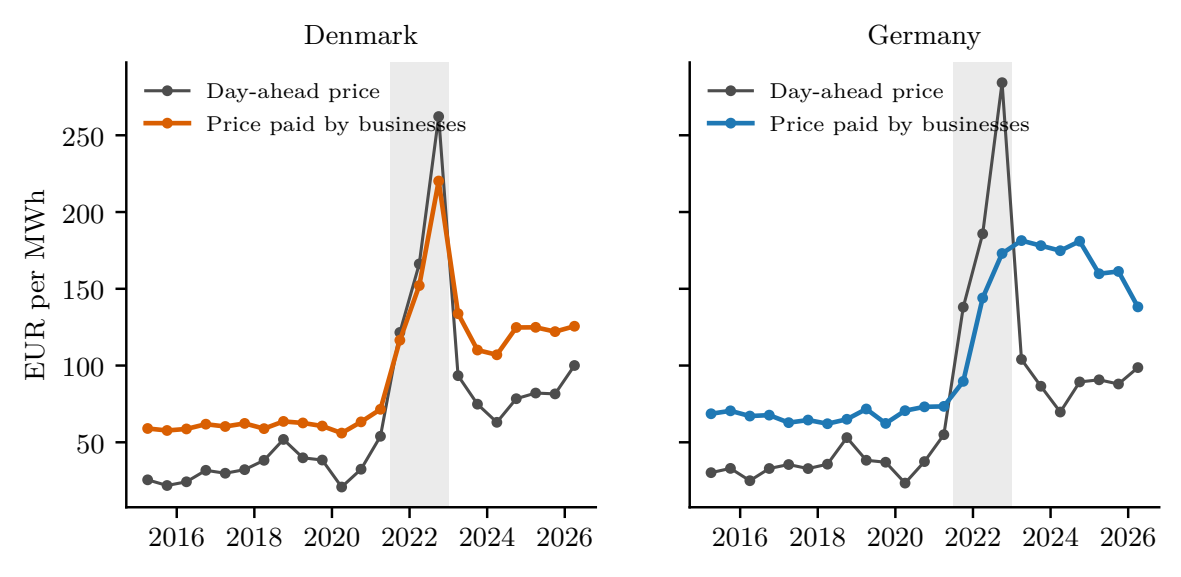


Convex demand: the same price difference gives a smaller consumption gap at higher prices.

Source: Ambec and Crampes (2021), eqs. (30)–(33). $\delta_2(\beta)$ increases with β regardless of the shape of demand, eq. (33).

A third explanation: insurance against price risk

Fixed-price contracts insure against long periods of high prices. In the model, consumers are only averse to varying *quantities* (Lemma 2), not to a varying *bill*.



Source: Energi Data Service; Eurostat (nrg_pc_205), 2,000–19,999 MWh/year, excl. taxes and VAT.

Q: What does this tell us about Danish and German contracts?

Insurance and incentives can be separated

A fixed-price contract does two things at once:

- It insures against the price *level* (e.g. 2022).
- It removes the incentive to shift consumption between *hours*.

The model only compares fixed prices with RTP, so it cannot tell the two apart. But they can be separated:

- Fix the price for an expected quantity, and pay the hourly price on deviations from it: a fixed-quantity forward contract (Borenstein, 2007).
- The consumer is insured, and the marginal price still signals scarcity.

What matters for flexibility is the *marginal* price signal, not the average price.